

Homework Submission

Student Name
September 5, 2026

Problem Statement

Solve the quadratic equation

$$x^2 + 4x + 4 = 0.$$

Key Principles and Formulas

- **Factoring (perfect square trinomial):** If a quadratic can be written as $(x+p)^2 = 0$, then the only solution is $x = -p$.
- **Quadratic formula:** For $ax^2 + bx + c = 0$ with $a \neq 0$, the roots are

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

The discriminant $\Delta = b^2 - 4ac$ determines the nature of the roots: $\Delta > 0$ two distinct real roots, $\Delta = 0$ one repeated real root, $\Delta < 0$ two complex conjugate roots.

- **Completing the square:** Rewrite $ax^2 + bx + c$ as $a\left(x + \frac{b}{2a}\right)^2 + \text{constant}$ to solve for x .

Detailed Step-by-Step Solution

Method 1: Factoring as a perfect square

$$\begin{aligned}x^2 + 4x + 4 &= 0 \\x^2 + 4x + 4 &= (x + 2)^2 \\(x + 2)^2 &= 0 \\x + 2 &= 0 \\x &= -2\end{aligned}$$

Method 2: Quadratic formula

$$\begin{aligned}a &= 1, & b &= 4, & c &= 4 \\ \Delta &= b^2 - 4ac = 4^2 - 4 \cdot 1 \cdot 4 = 16 - 16 = 0 \\ x &= \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{-4 \pm \sqrt{0}}{2 \cdot 1} = \frac{-4 \pm 0}{2} \\ x &= -2\end{aligned}$$

Both methods give the same root.

Final Answer

$$\boxed{x = -2}$$

Study Notes and Key Takeaways

- A discriminant equal to zero ($\Delta = 0$) signals a *repeated* real root.
- Recognize the pattern $a^2 + 2ab + b^2 = (a + b)^2$; here $x^2 + 4x + 4 = (x + 2)^2$.
- Always verify the solution by substituting $x = -2$ back into the original equation:
 $(-2)^2 + 4(-2) + 4 = 0$.